

Multitask Deep Learning Framework for Medical Imaging

Introduction

Medical image quality significantly impacts clinical diagnostic accuracy and treatment planning.

The unique characteristics of medical images, such as including their high dynamic range of an image, semantic movement or implied movement, specific structural patterns, and boundary of critical diagnostic regions demand specialized approaches that differ from conventional traditional image processing methods.

Two major projects :

- **Super-Resolution Project**
- **Pneumonia Classification Project**

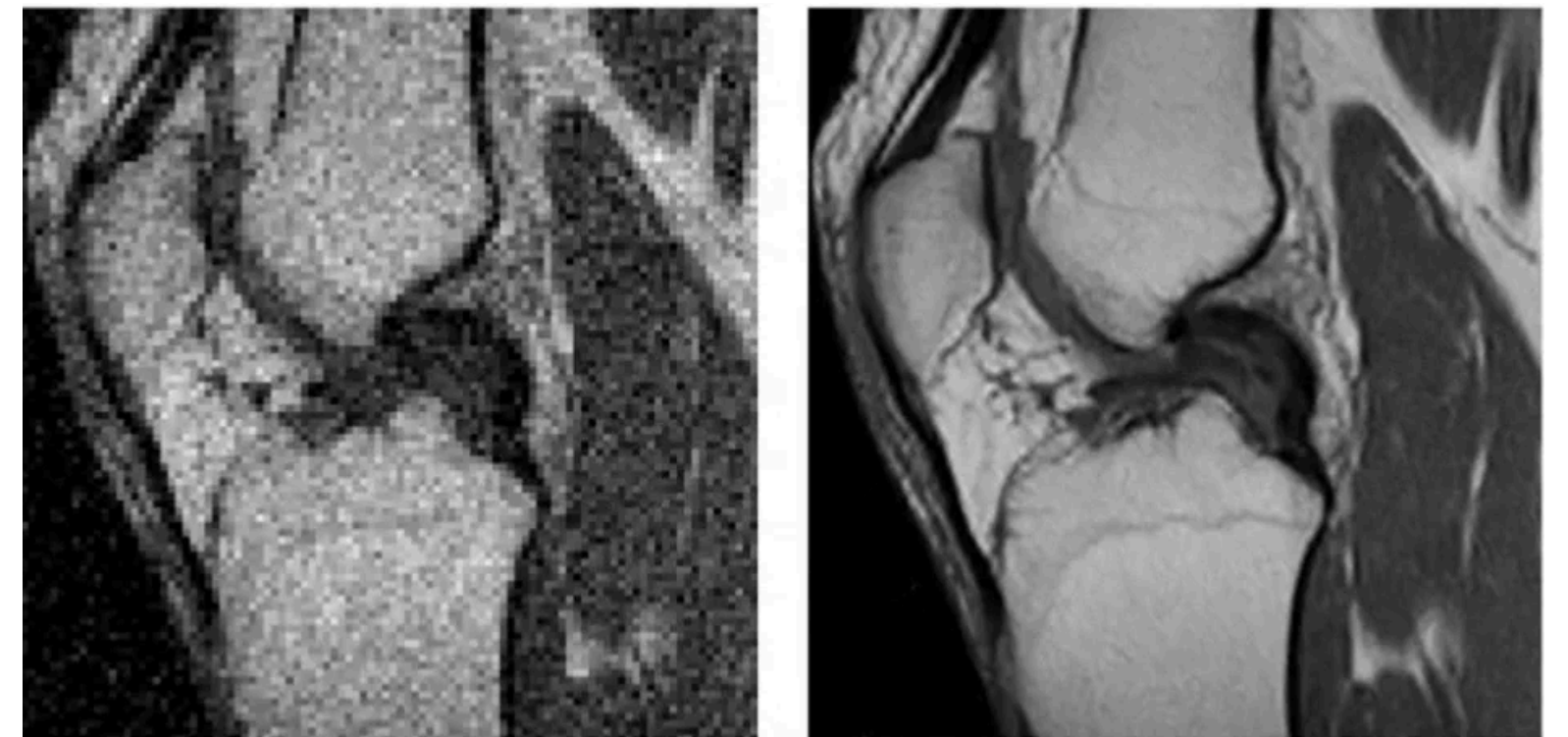


Image super-resolution

Computed tomography (CT), X-ray imaging is the first technology to generate 2D or 3D images of the internal structure of the human body.

Problems with CT imaging and X-ray imaging are health concerns for X-ray radiation. It is desirable to achieve clinically acceptable image quality while lowering radiation dose as much as possible.

The image quality is related to many factors in the reconstruction process, such as **detector characteristics, patient motions, artifacts, computational errors, radiation dose, etc.**

Thus, to obtain high-quality CT images, noise cancellation(denoising) and deblurring can be transformed into a super-resolution problem that aims to restore bad images that have been pre-degraded to optimal high-resolution images.

Challenges to solve

Computational cost burden is exacerbated when processing large volumes of medical images, as is common in clinical settings.

A typical high-resolution medical image can require several GBs(gigabytes) of memory for processing, making real-time or batch processing impractical without specialized hardware.

Clinical reliability is still the most important concern.

The generation of super-resolved medical images must maintain absolute fidelity to the original diagnostic information.

Any artifacts introduced during the enhancement process could lead to misdiagnosis or incorrect treatment decisions.

The method we will introduce this time was to solve this problem by integrating tile processing algorithms and diffusion models with improved performance.

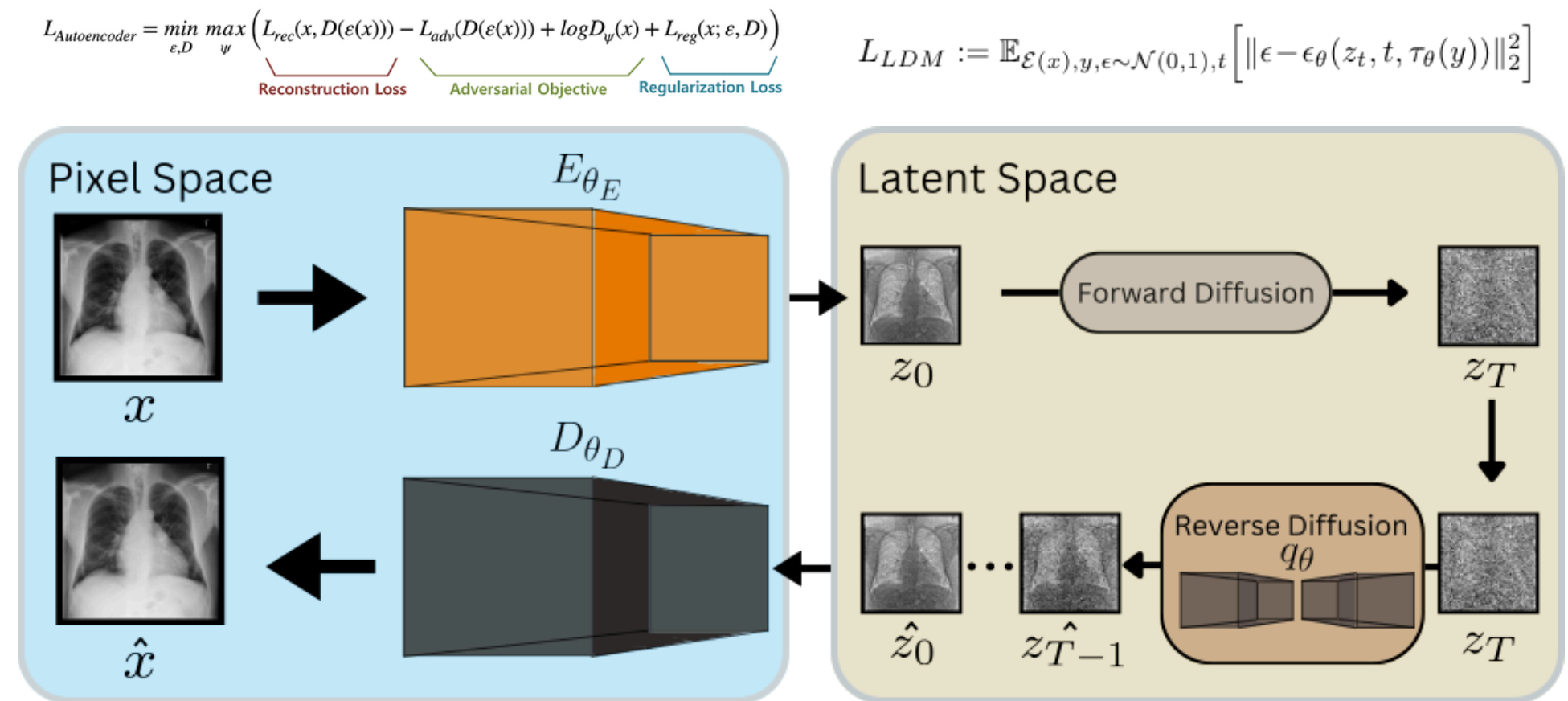
Latent Diffusion Models

The Latent Diffusion Model is one of the deep generative models that generates data using a **forward diffusion process** that gradually adds noise in a low dimensional latent space to make data complete noise and a **reverse process** that generates data by gradually restoring it from noise.

This is because the Latent Diffusion model is designed to learn the distribution of data using VAE in **encoder-decoder** format.

Disadvantage is a computational cost for high-resolution image synthesis and takes a long time to perform the Diffusion process.

To address this, we have introduced an **adaptive tile processing algorithm**.

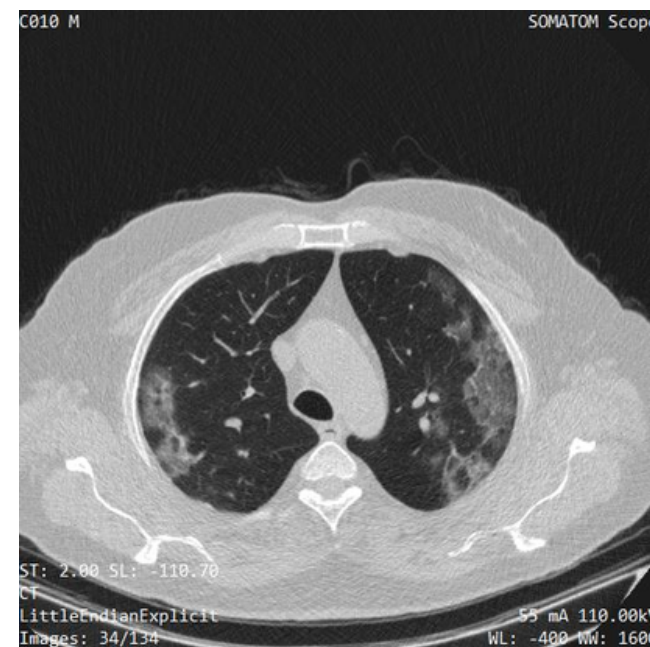
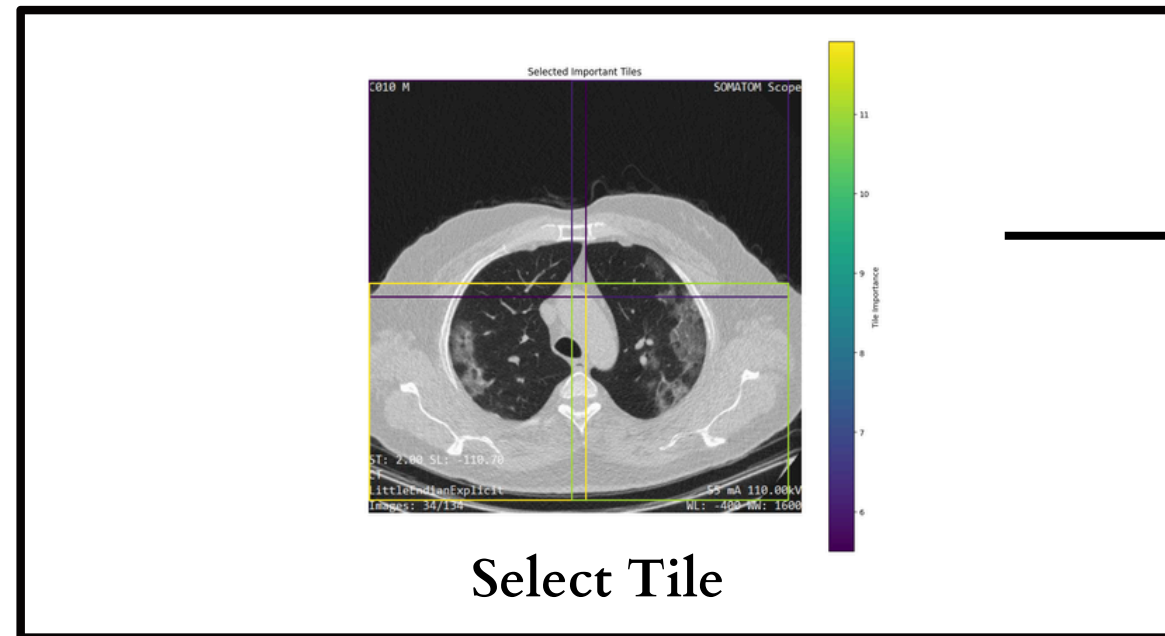


Latent Diffusion Models : Encoder projects data onto its low dimensional latent representation and followed by a back projection onto the original pixel space.

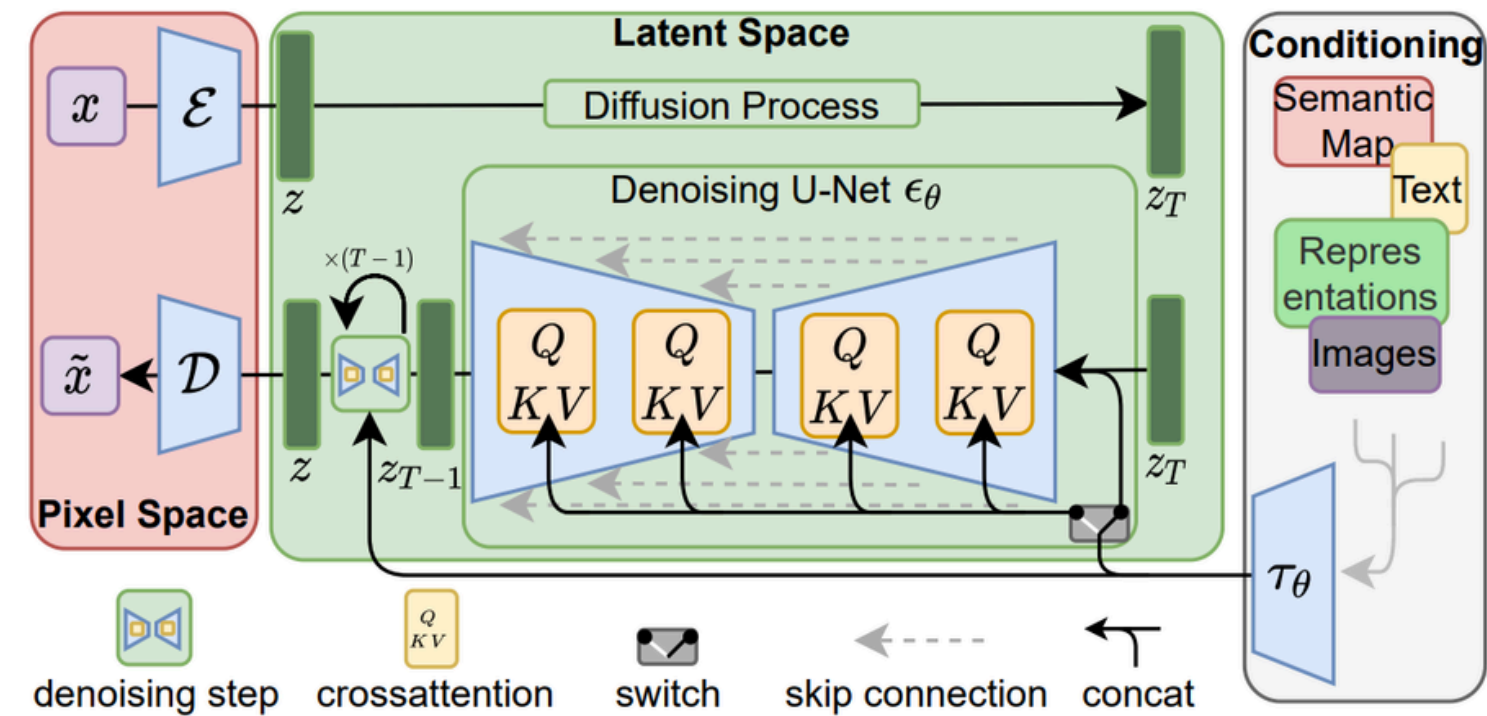
Adaptive Tile Processing algorithm



Low resolution CT images



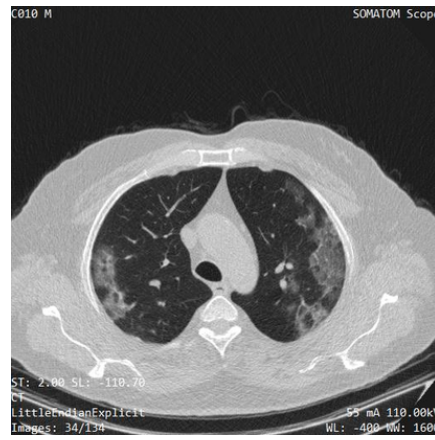
Upscaled CT images



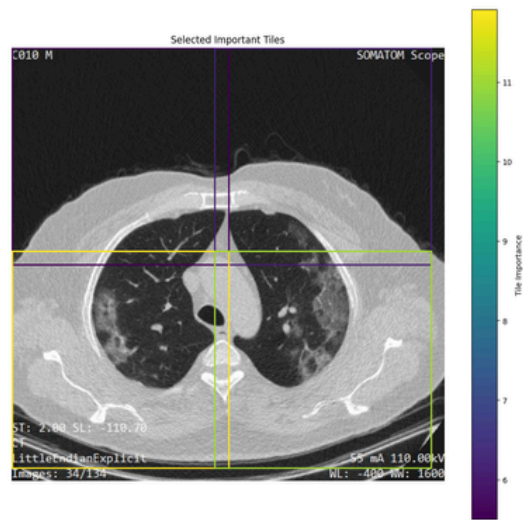
Pretrained LDM upscaler

Conditioning mechanism :
Text, Image

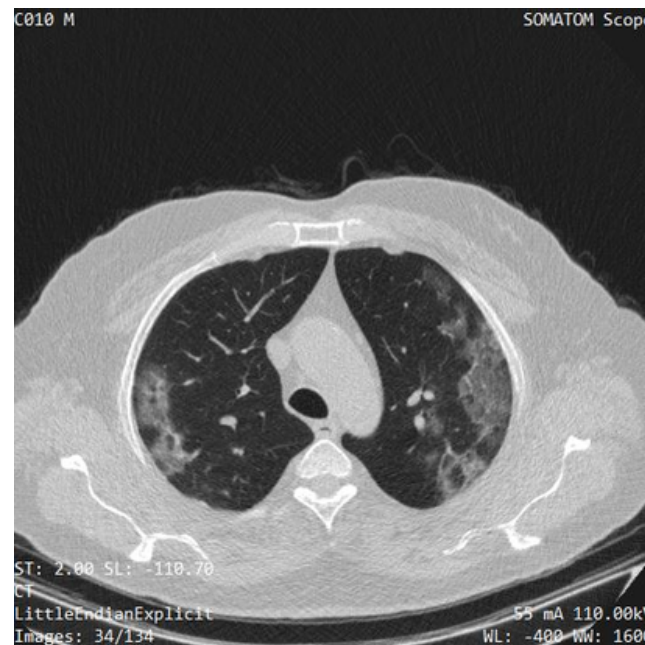
Fine Tuned Latent Diffusion Model



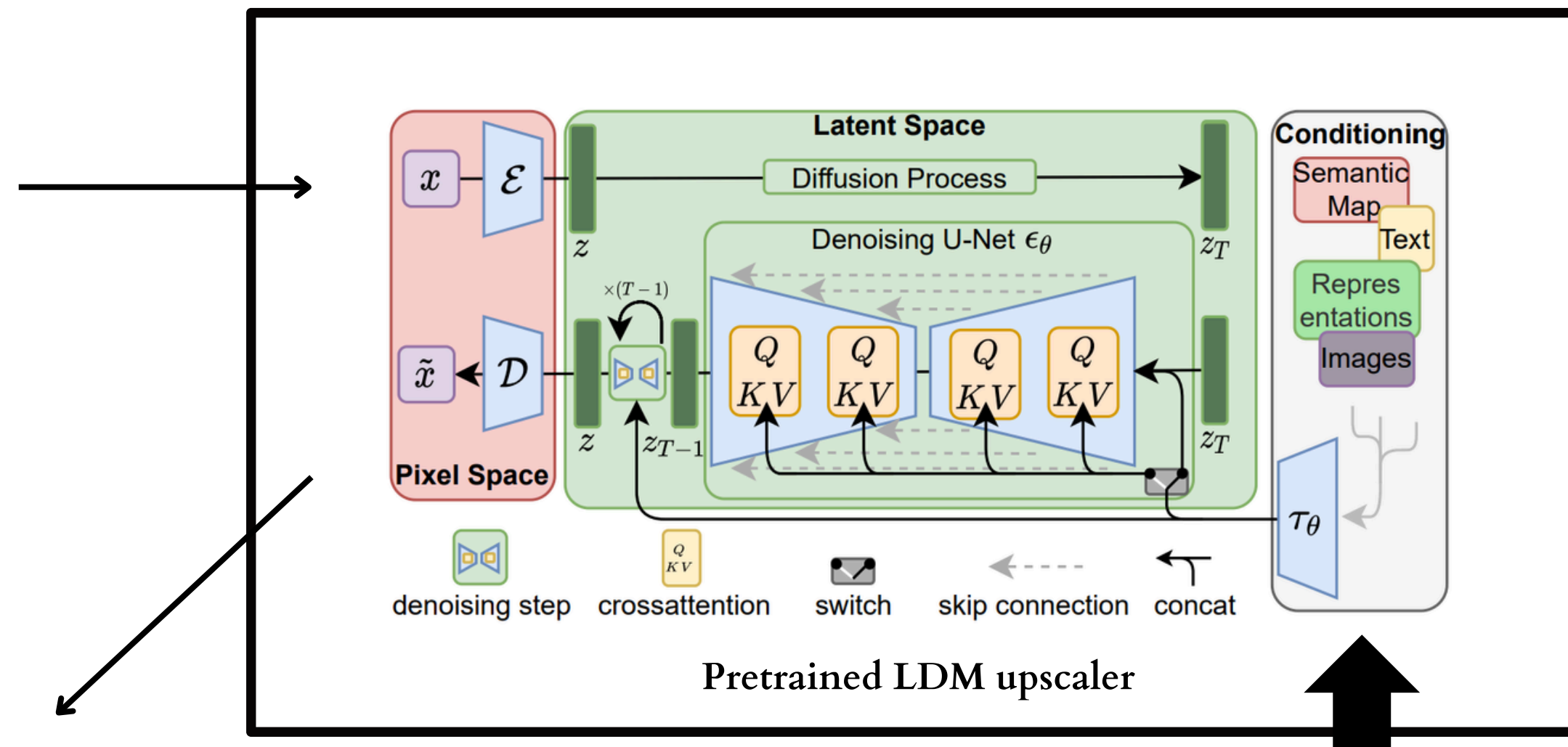
Low resolution CT images



Select Tile



Upscaled CT images



Conditioning
mechanism :
Text, Image

Used Dataset : Chest X-Ray Images (Pneumonia)

- 5,863 X-Ray images (JPEG) and 2 categories (**Pneumonia/Normal**).
- **Chest X-ray images** (anterior-posterior) were selected from retrospective cohorts of pediatric patients of one to five years old from Guangzhou Women and Children's Medical Center, Guangzhou.
- The normal chest X-ray (left panel) depicts clear lungs without any areas of abnormal opacification in the image.
- Bacterial pneumonia (middle) typically exhibits a focal lobar consolidation, in this case in the right upper lobe (white arrows), whereas viral pneumonia (right) manifests with a more diffuse “interstitial” pattern in both lungs.



Illustrative Examples of Chest X-Rays in Patients with Pneumonia

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Identifying Medical Diagnoses and Treatable Diseases by Image-Based De...

Image-based deep learning classifies macular degeneration and diabetic retinopathy using retin...

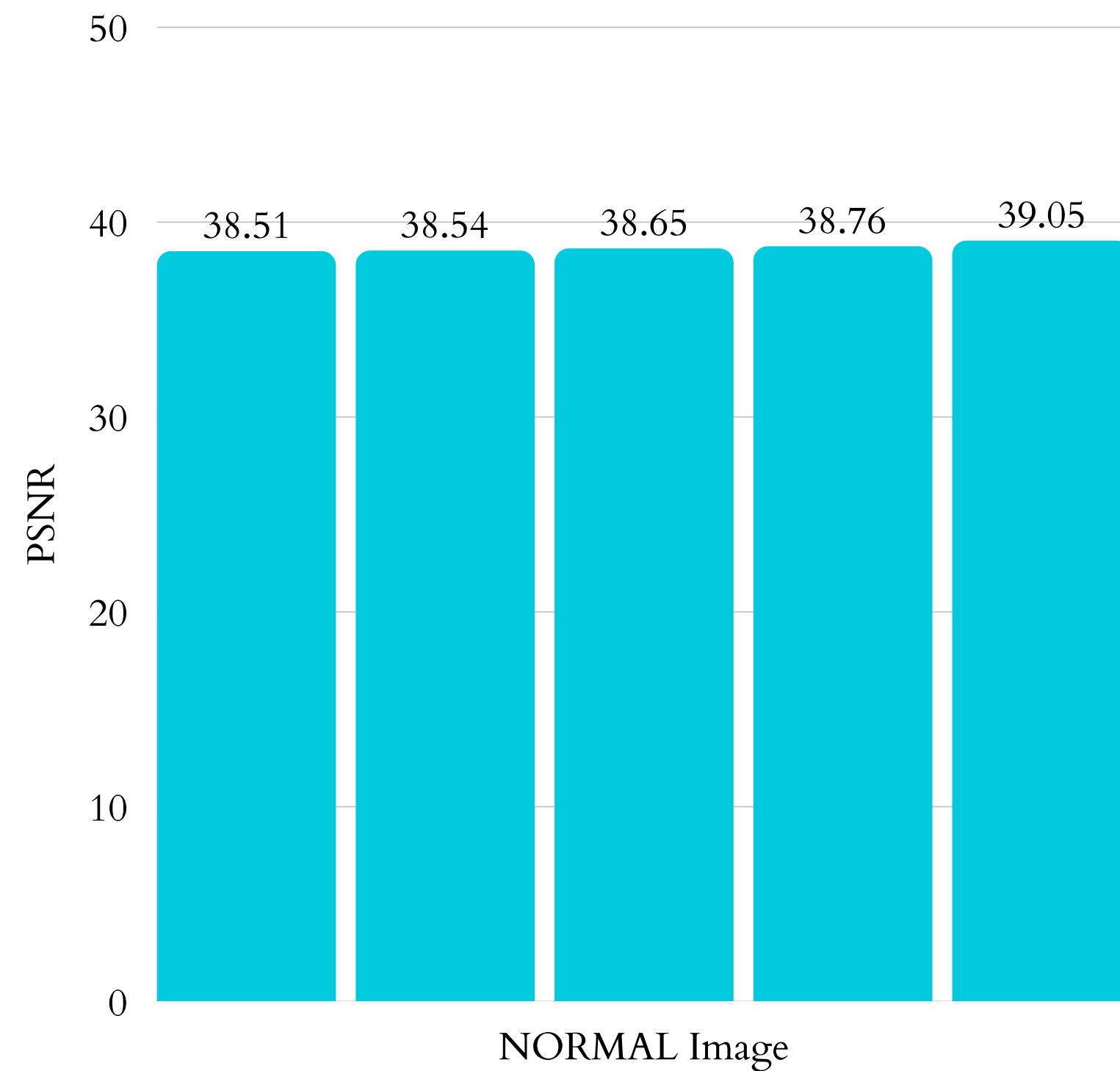
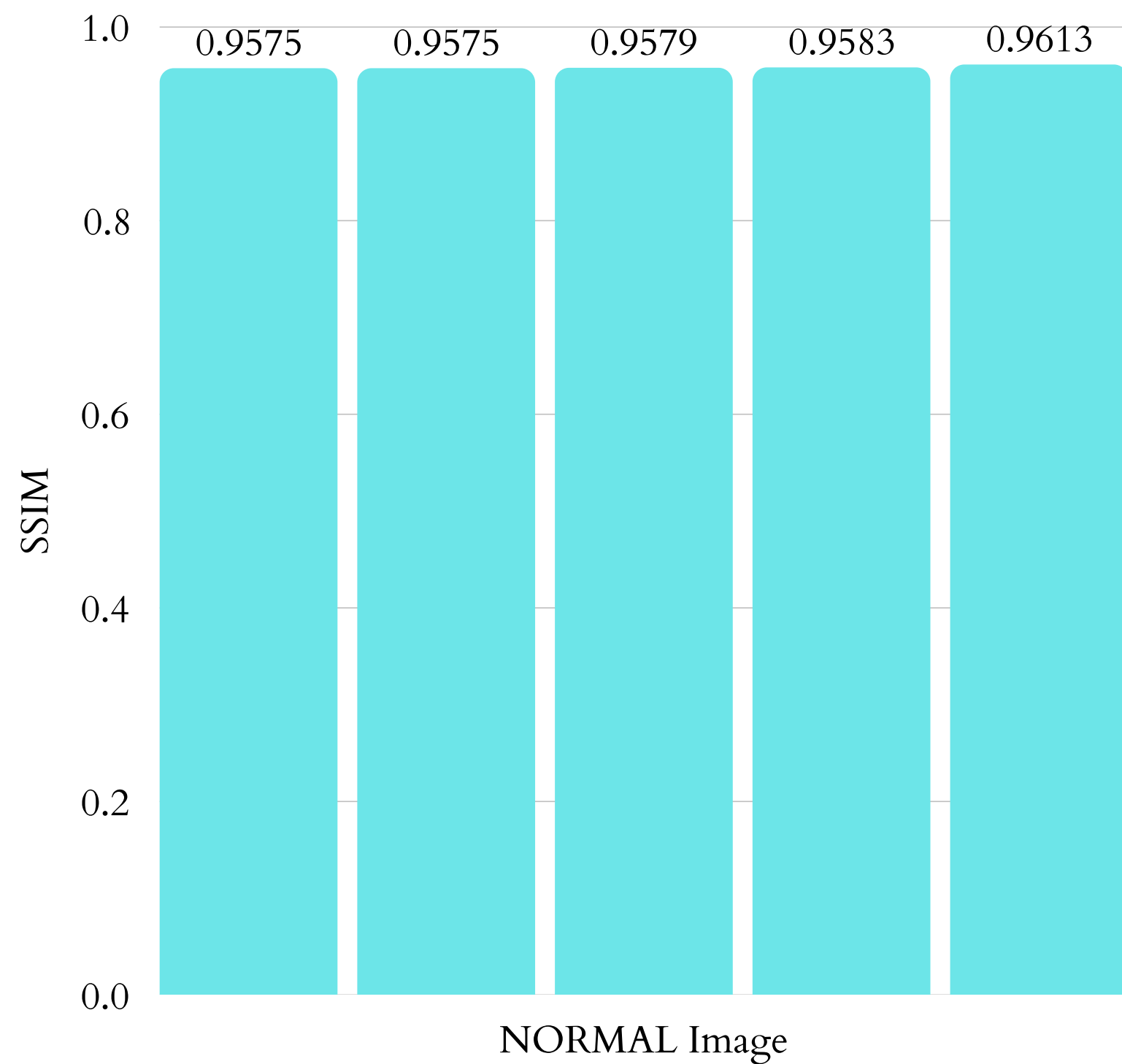
[Cell](#)

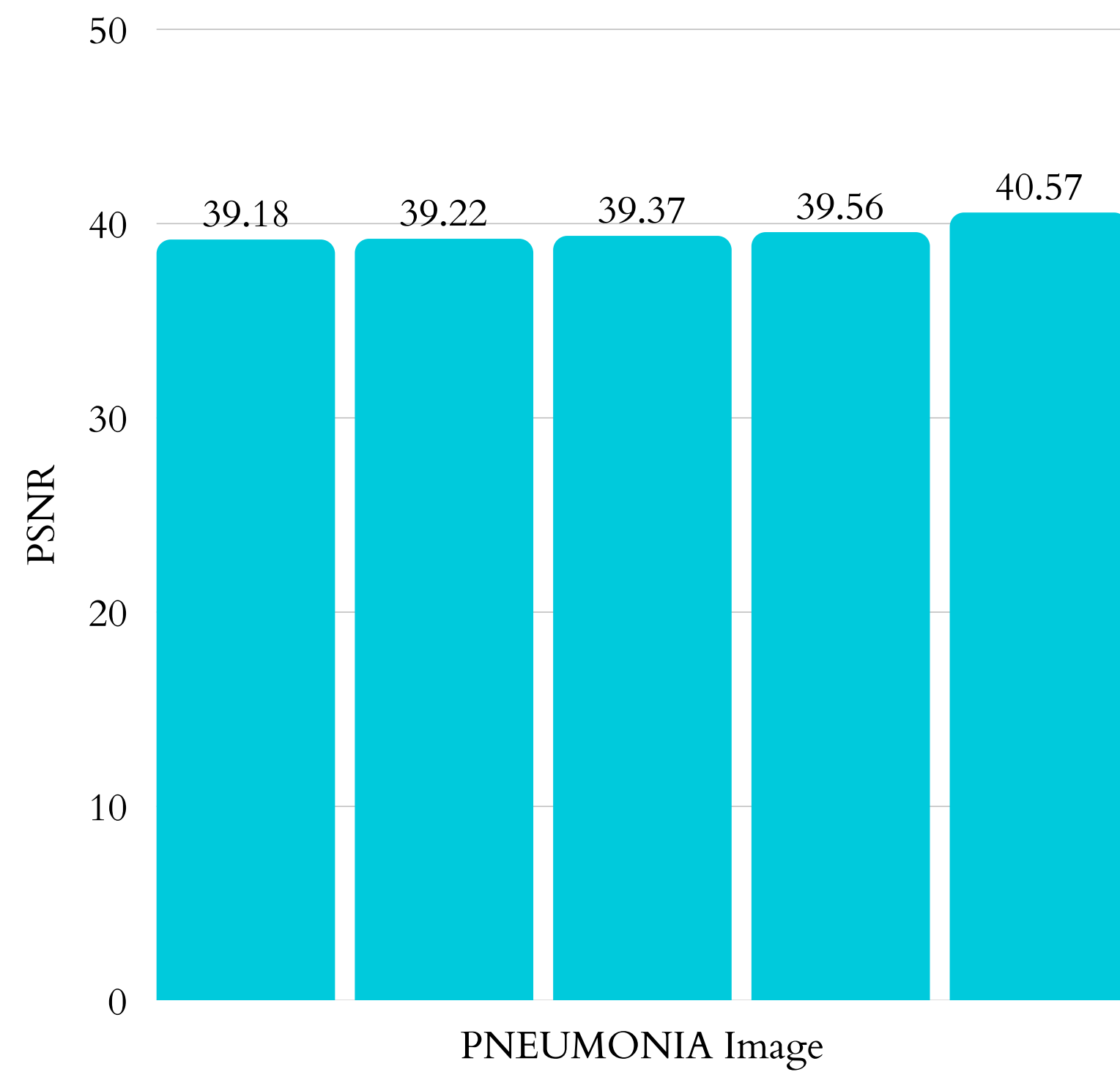
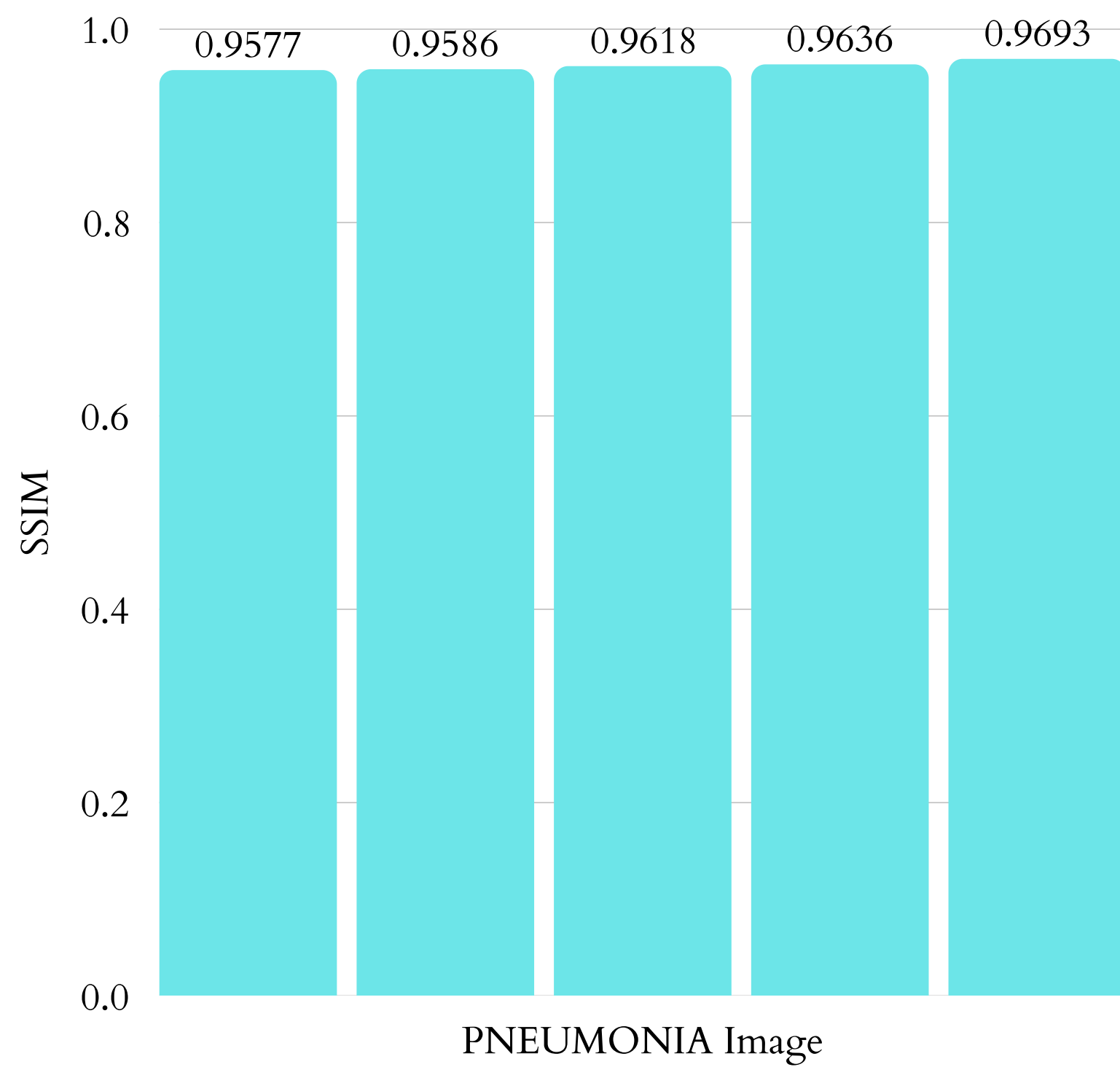
Labeled Optical Coherence Tomography (OCT) and Chest X-Ray Images for Classification

Dataset of validated OCT and Chest X-Ray images described and analyzed in "Deep learning-based classification and referral of treatable human diseases". The OCT Images are split into a training set and a...

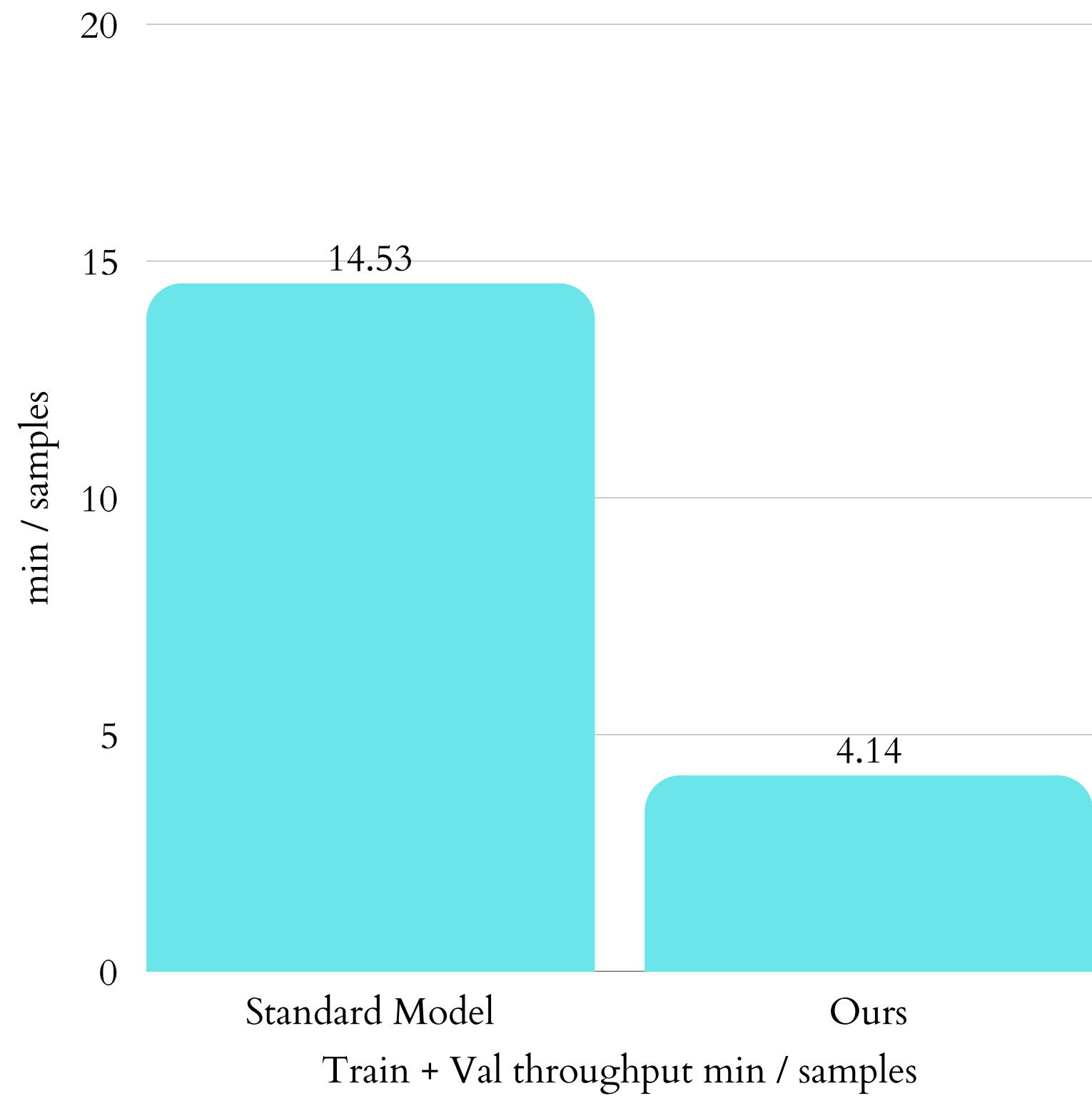
[Mendeley Data](#) / Jan 6, 2018

Super-Resolution Result





Time to process the image (512*512)

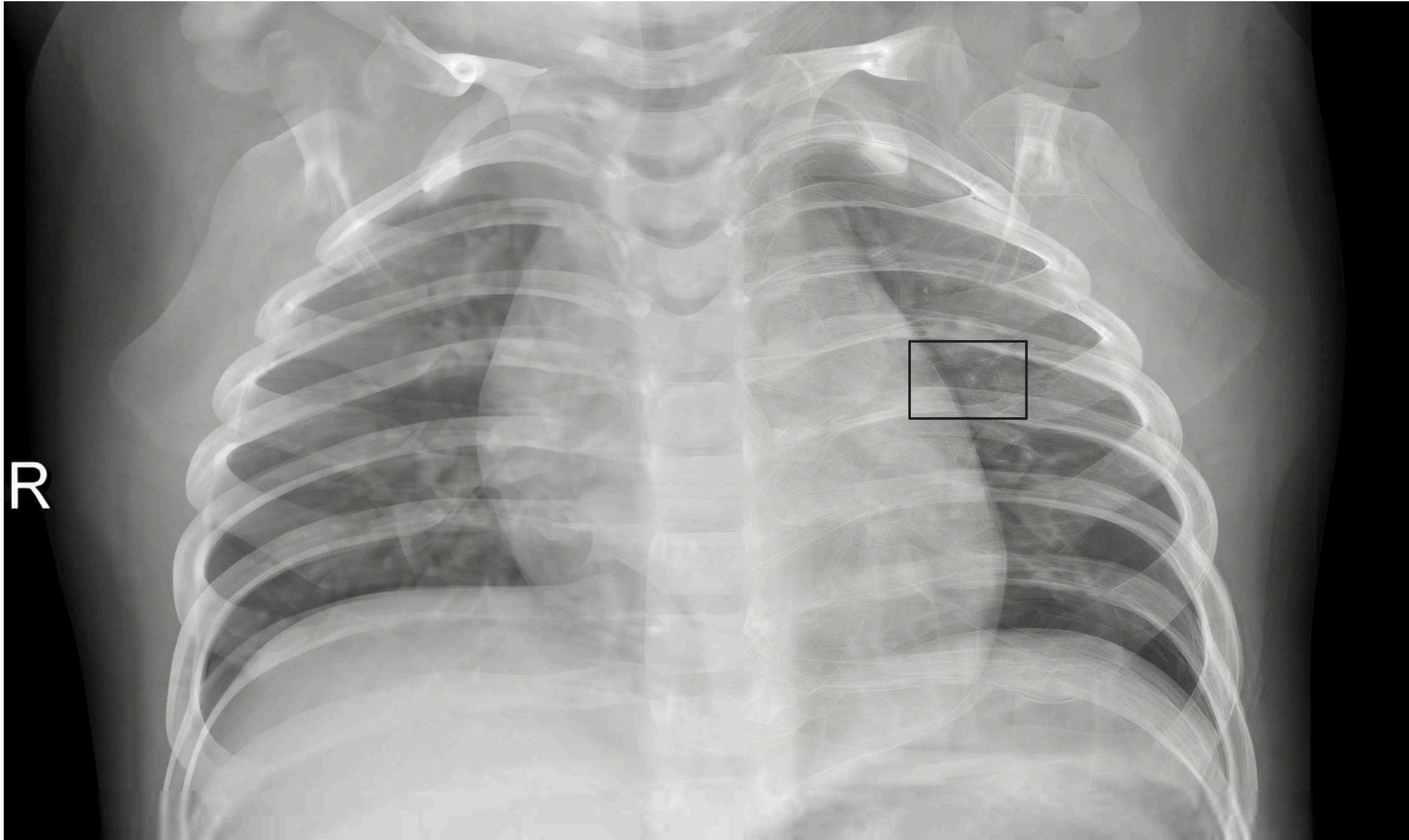


Assessed on a Tesla T4

Super-Resolution Experiments : Magnification

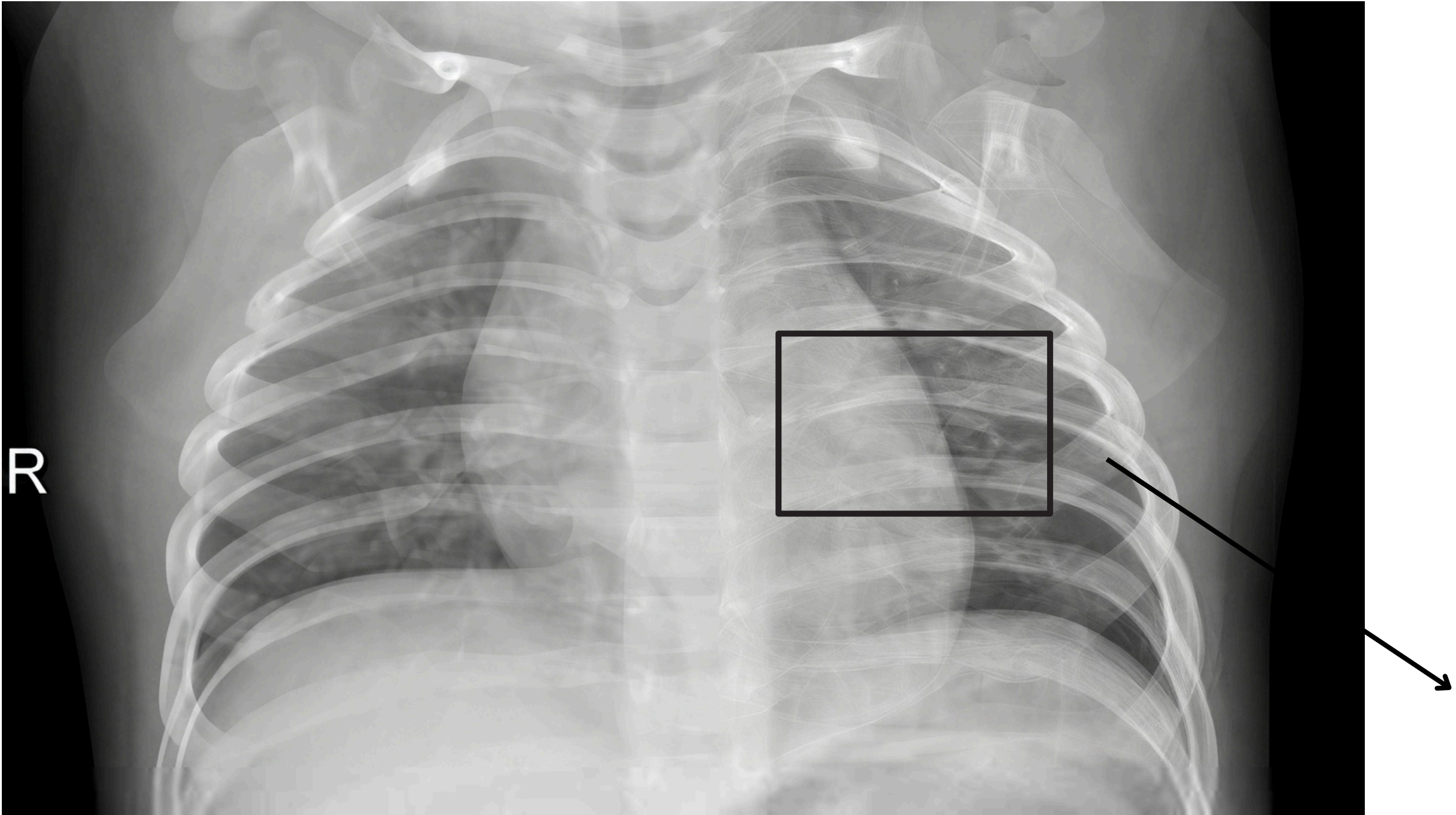


NORMAL2-IM-1141-0001.jpeg

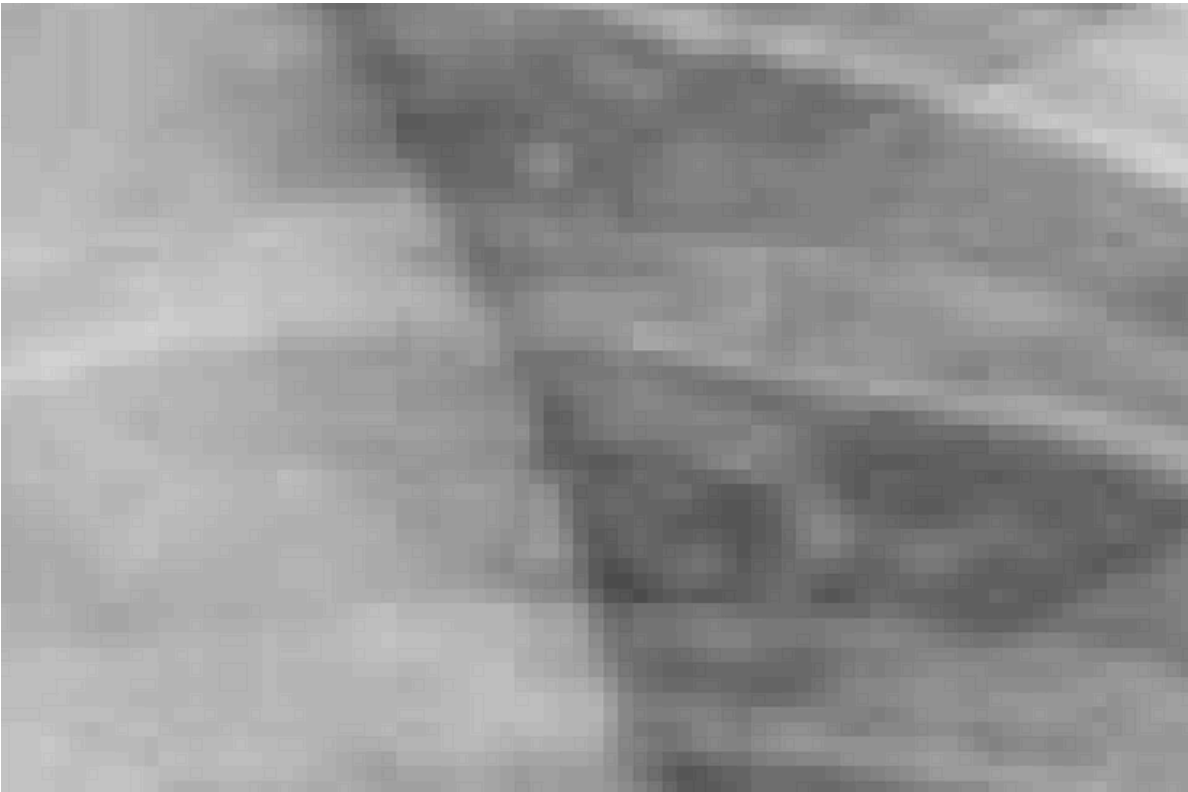


PSNR: 39.05 dB
SSIM: 0.9480

Super-Resolution Experiments : Magnification



Low resolution Image



Super-resolution Image

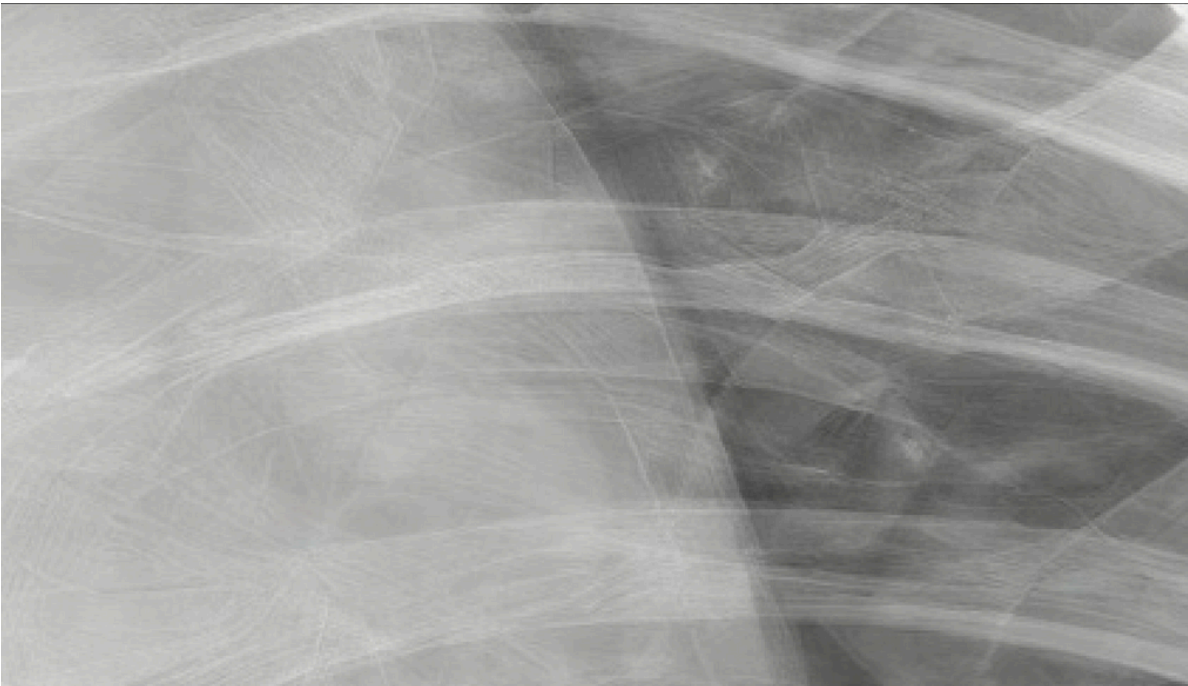
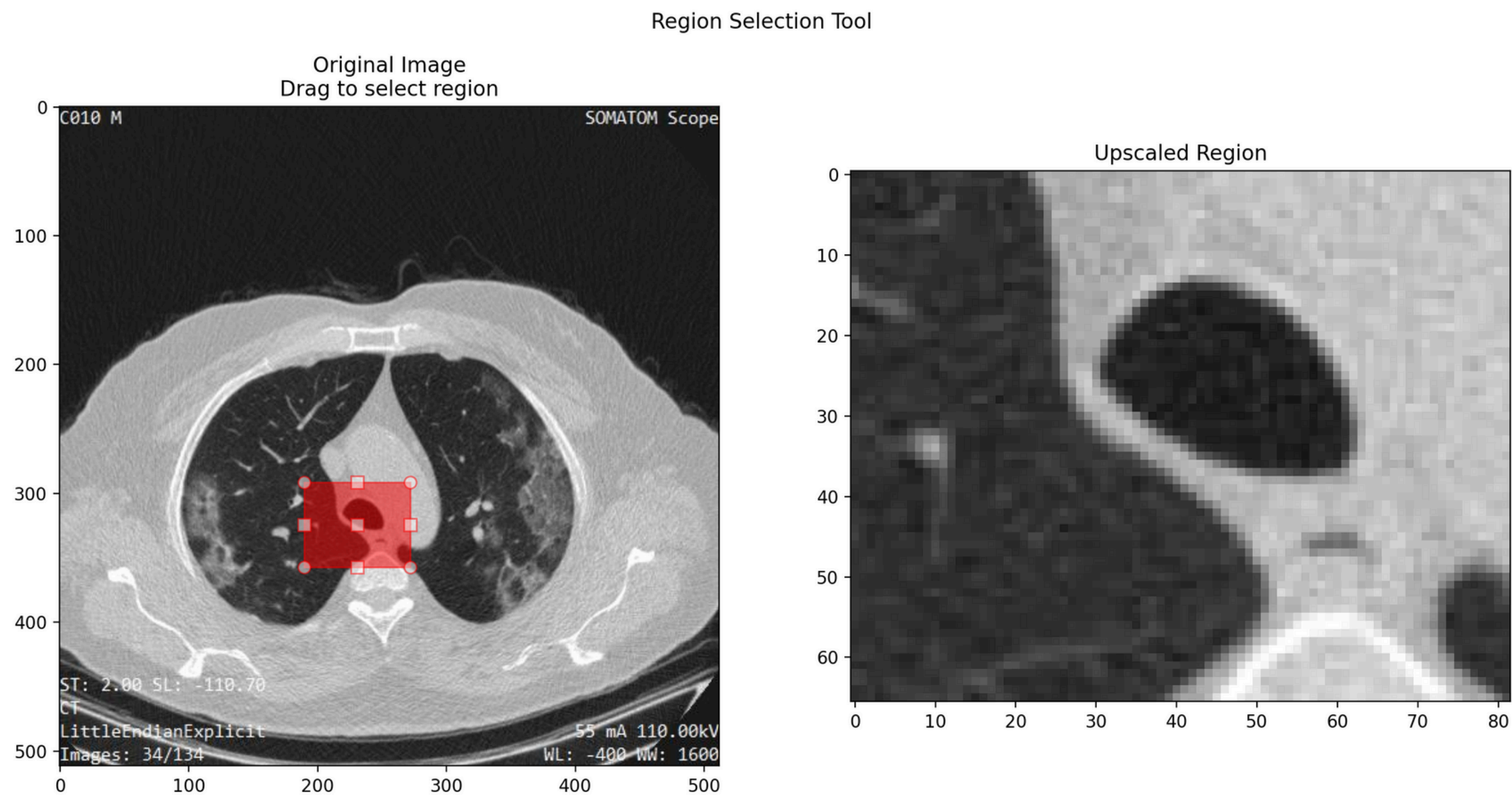
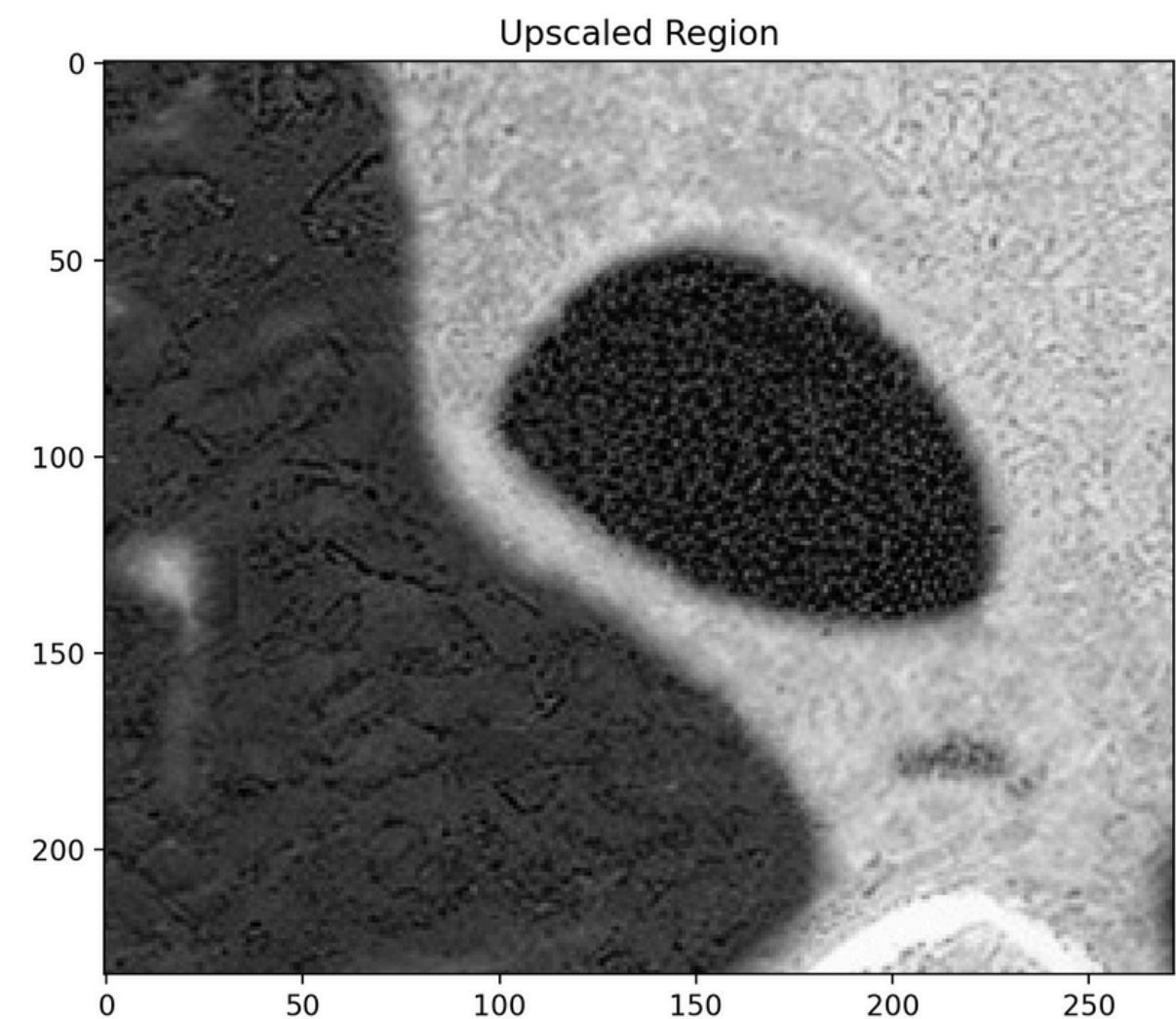
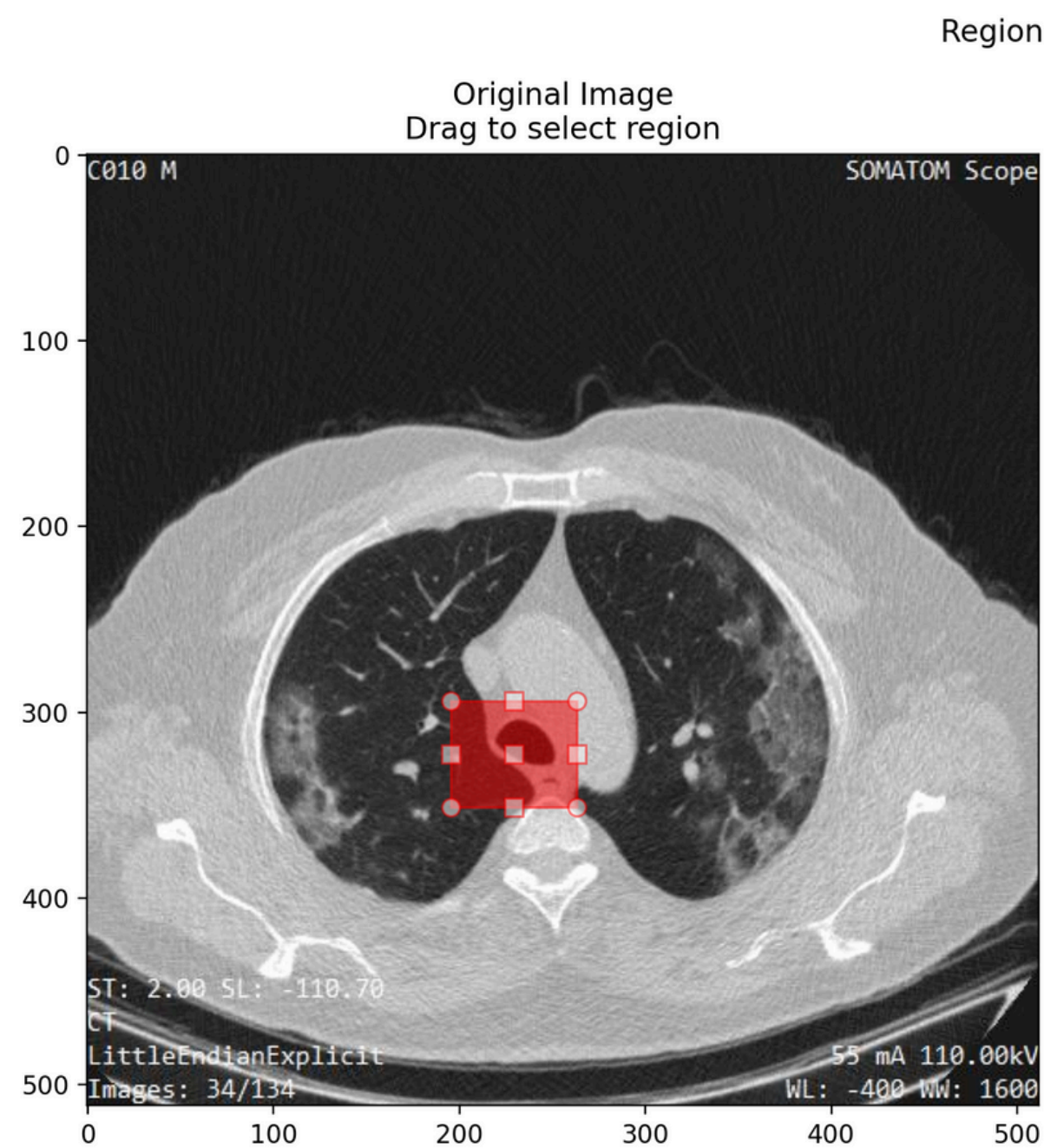


Image Magnification : CT Image



Experiments Results



Pneumonia Classification

Image Classification is primarily for binary classification, such as disease/cancer diagnostics.

The task of classifying diagnostic test images as diseased or **normal/abnormal**.

Pneumonia Classification is a task to diagnose pneumonia.

Using labeled X-ray image or CT image data, a deep learning model can be trained to determine whether pneumonia is present or not.

The data were based on the previously described Chest X-ray data.

Modified CNN

Medical image analysis often suffers from limited data. Pre-trained models can be used to detect or diagnose lesions in MRI or CT scan images. These approaches provide effective solutions when data is scarce in the medical field.

The model called Modified CNN is made by fine-tuning MovilenetV2.

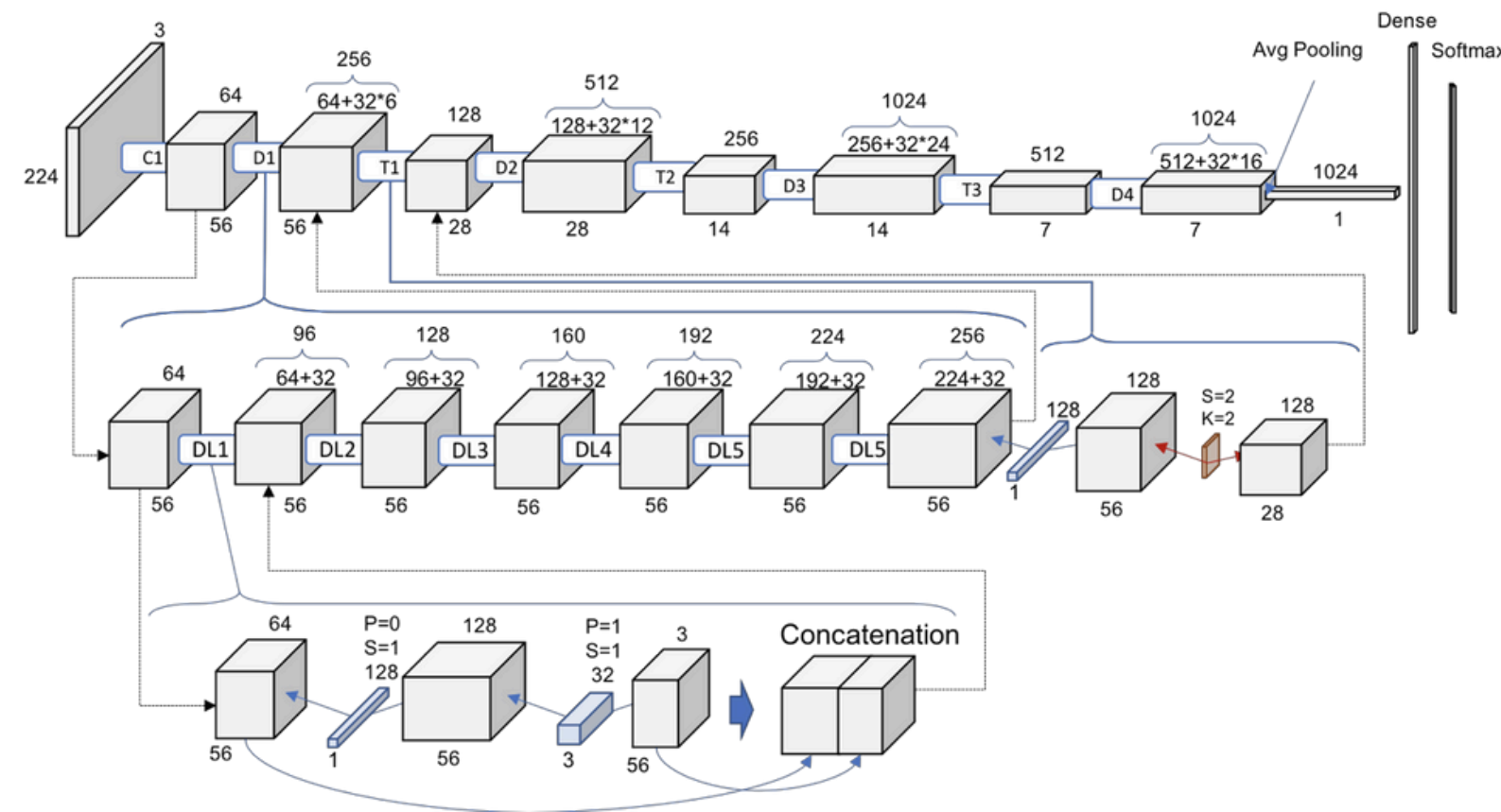
1. **Data Augmentation** : Use RandomBrightnessContrast, RandomGamma, CLAHE, ElasticTransform, HorizontalFlip, etc.
2. **Modified architecture** : Add Classifier Layers to Improve Generalization performance
3. **Optimizer** : adamW

Pneumonia Classification (Experiment)

The models used in the comparison and high-resolution verification experiments are the denenet121 model and the Resnet50 model.

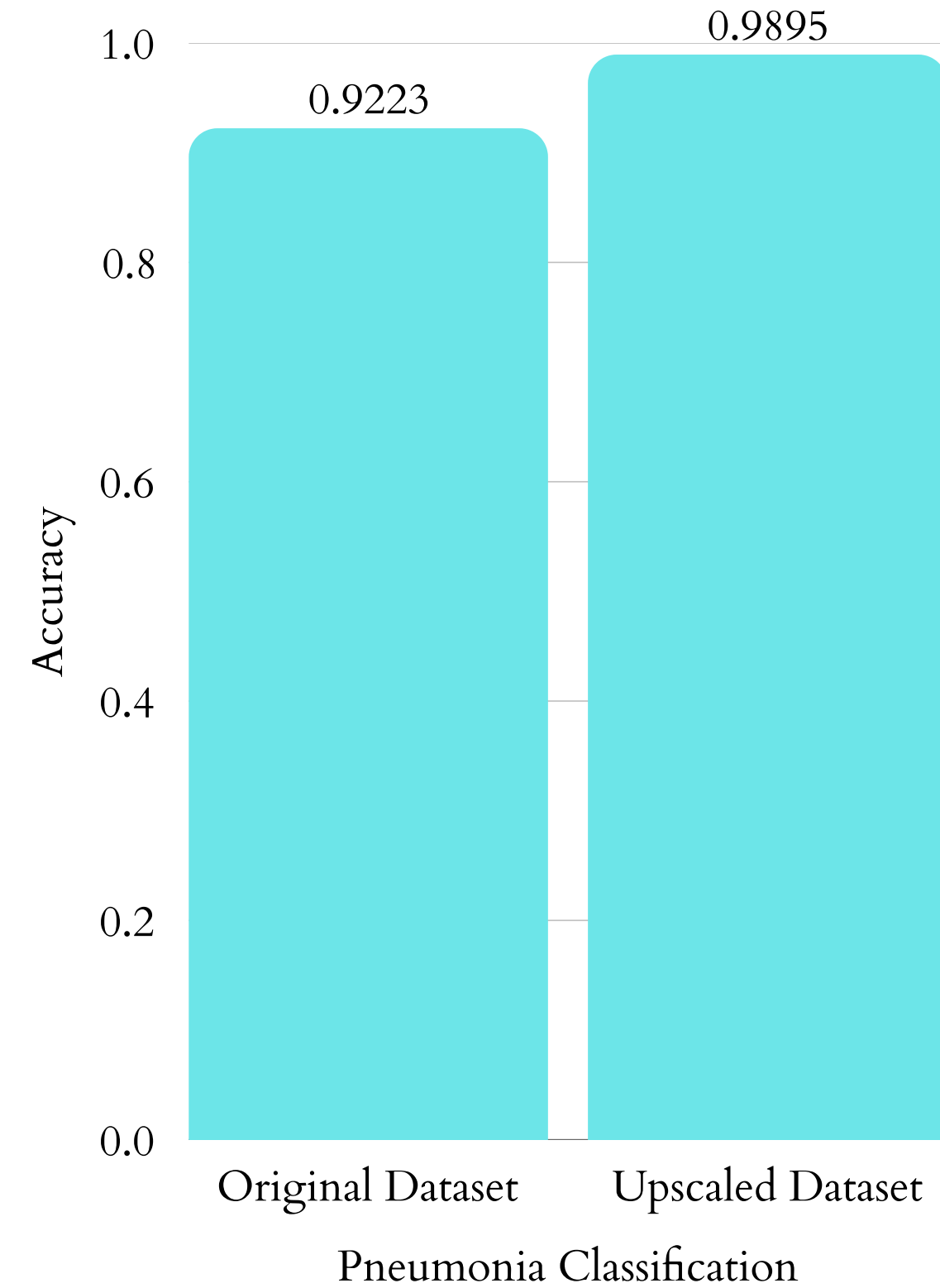
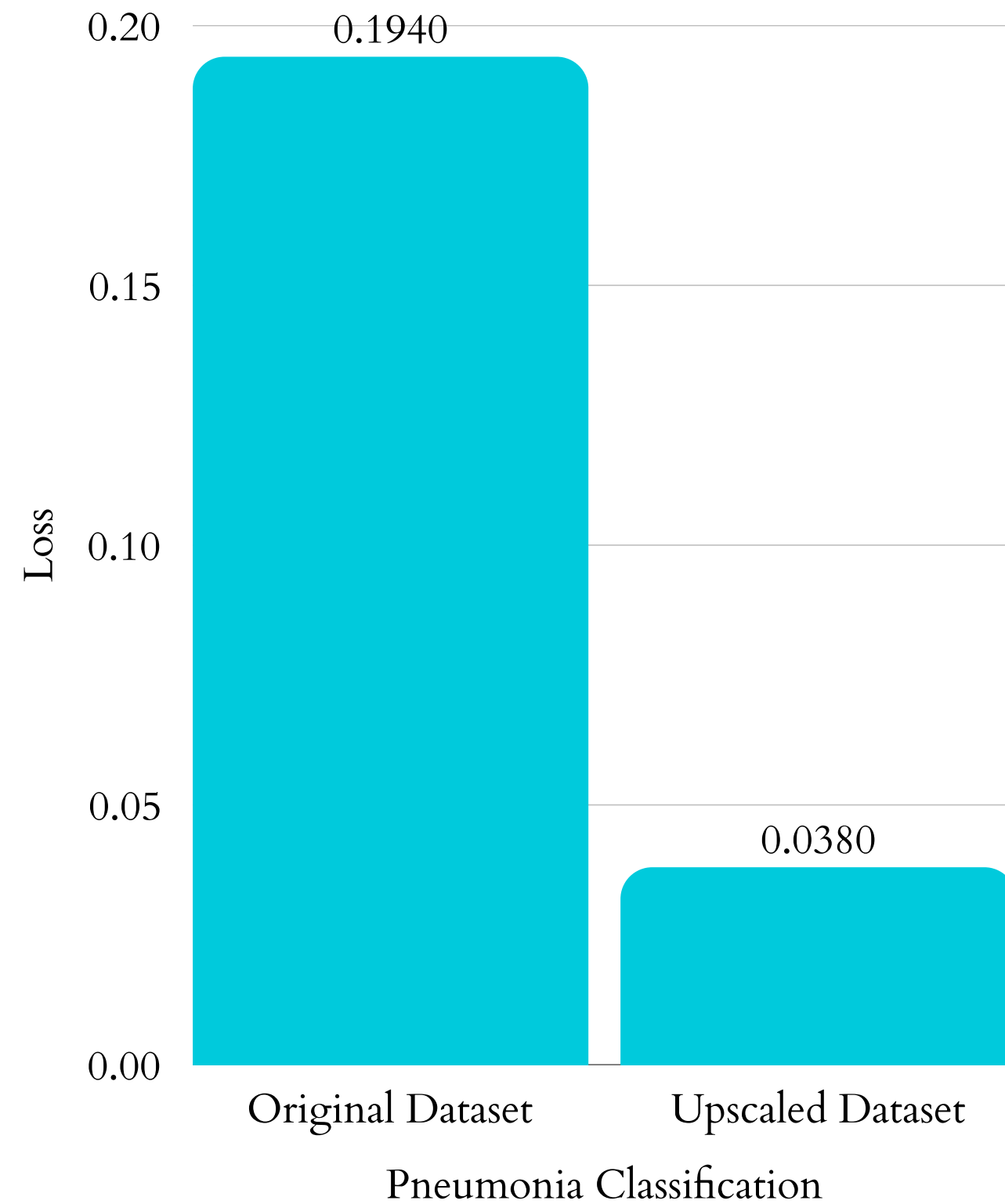
Densenet121

It belongs to the family of convolutional neural networks (CNNs) and is characterized by its unique connectivity pattern.

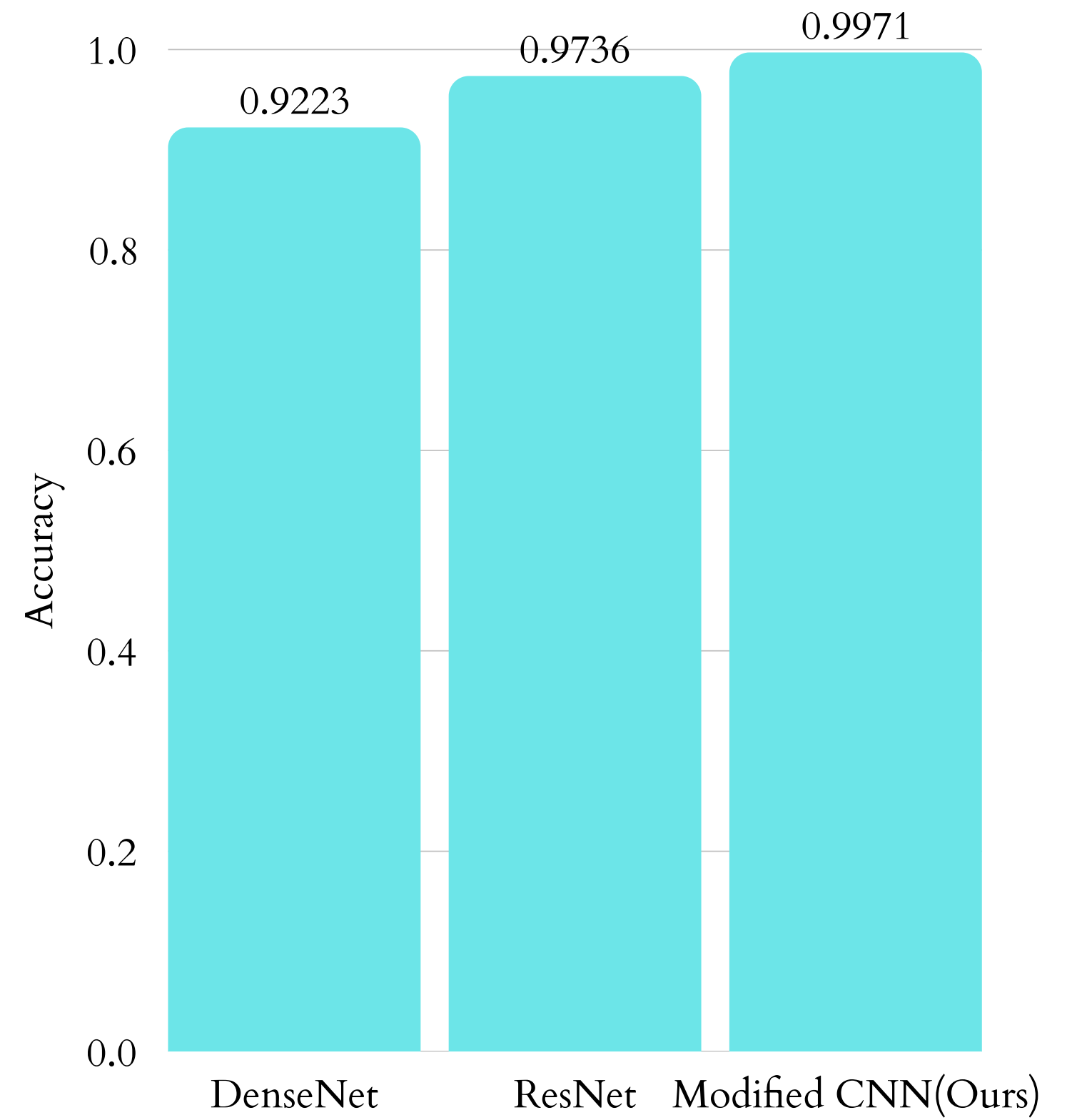
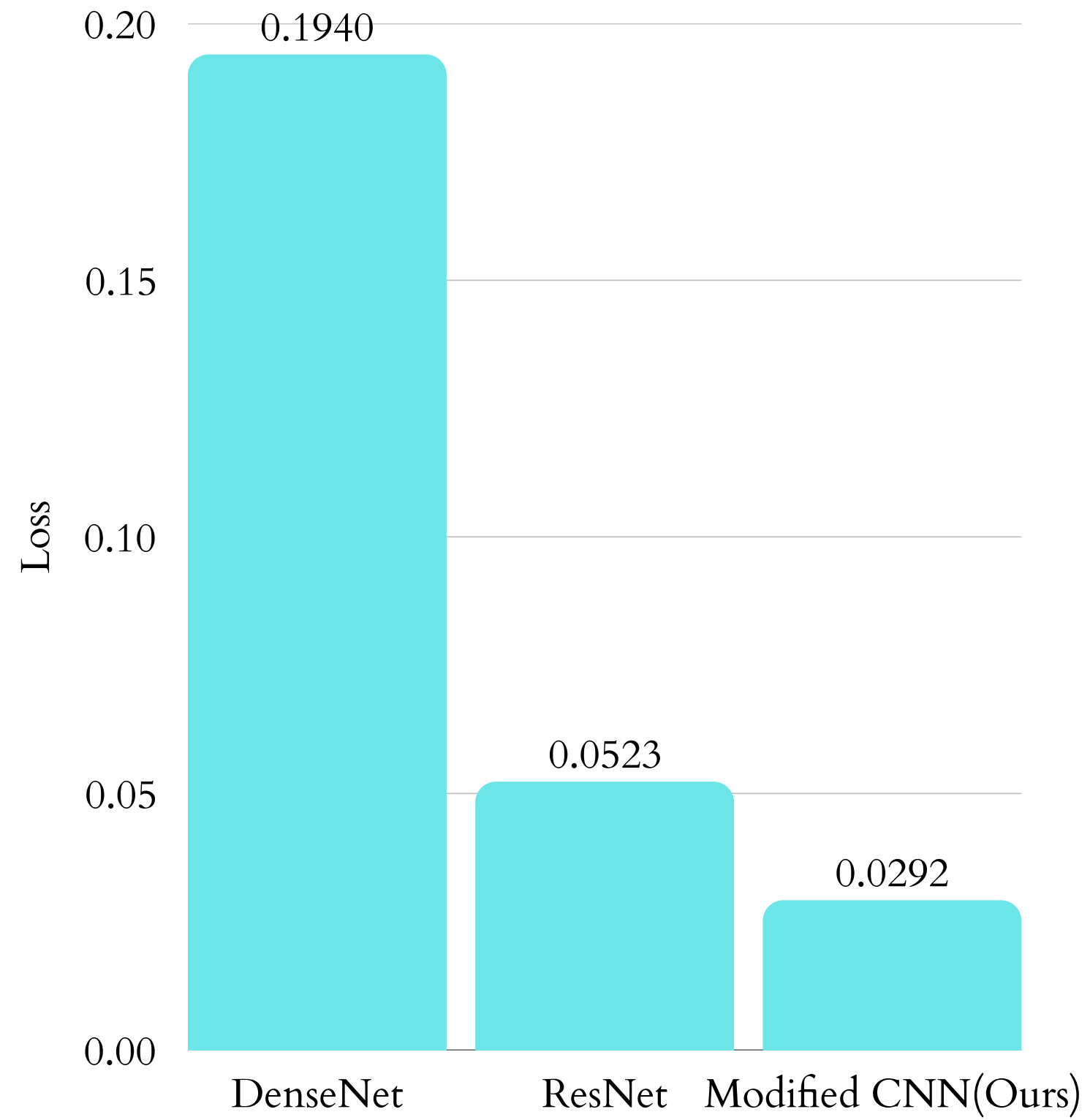


DenseNet-121. At feed forward, each layer connects with all other layers. There are four layers excluding input, and all of them are connected. DenseNet connects the previous layer directly to all the next layers.

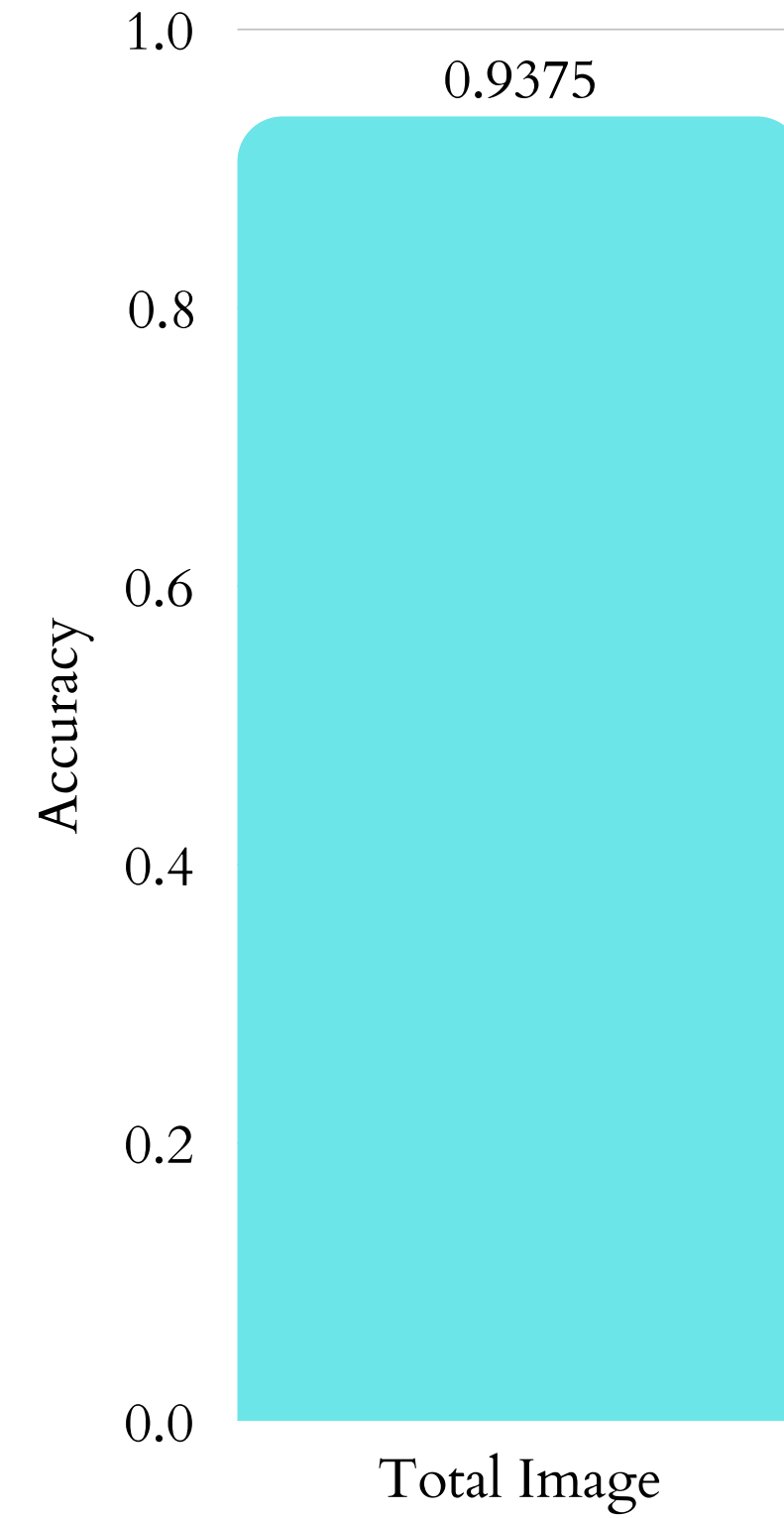
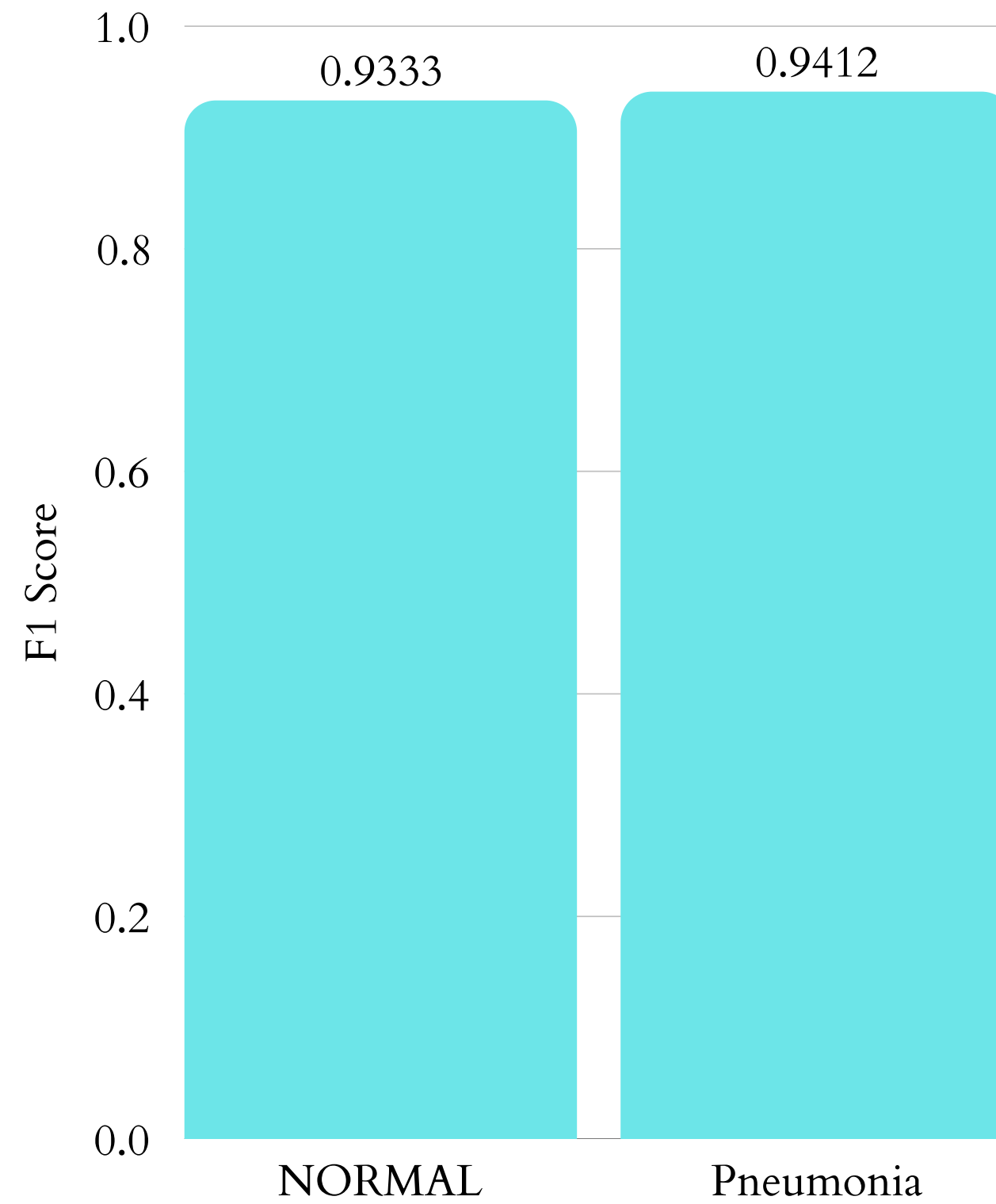
Classification Result (with Super-Resolution Image)



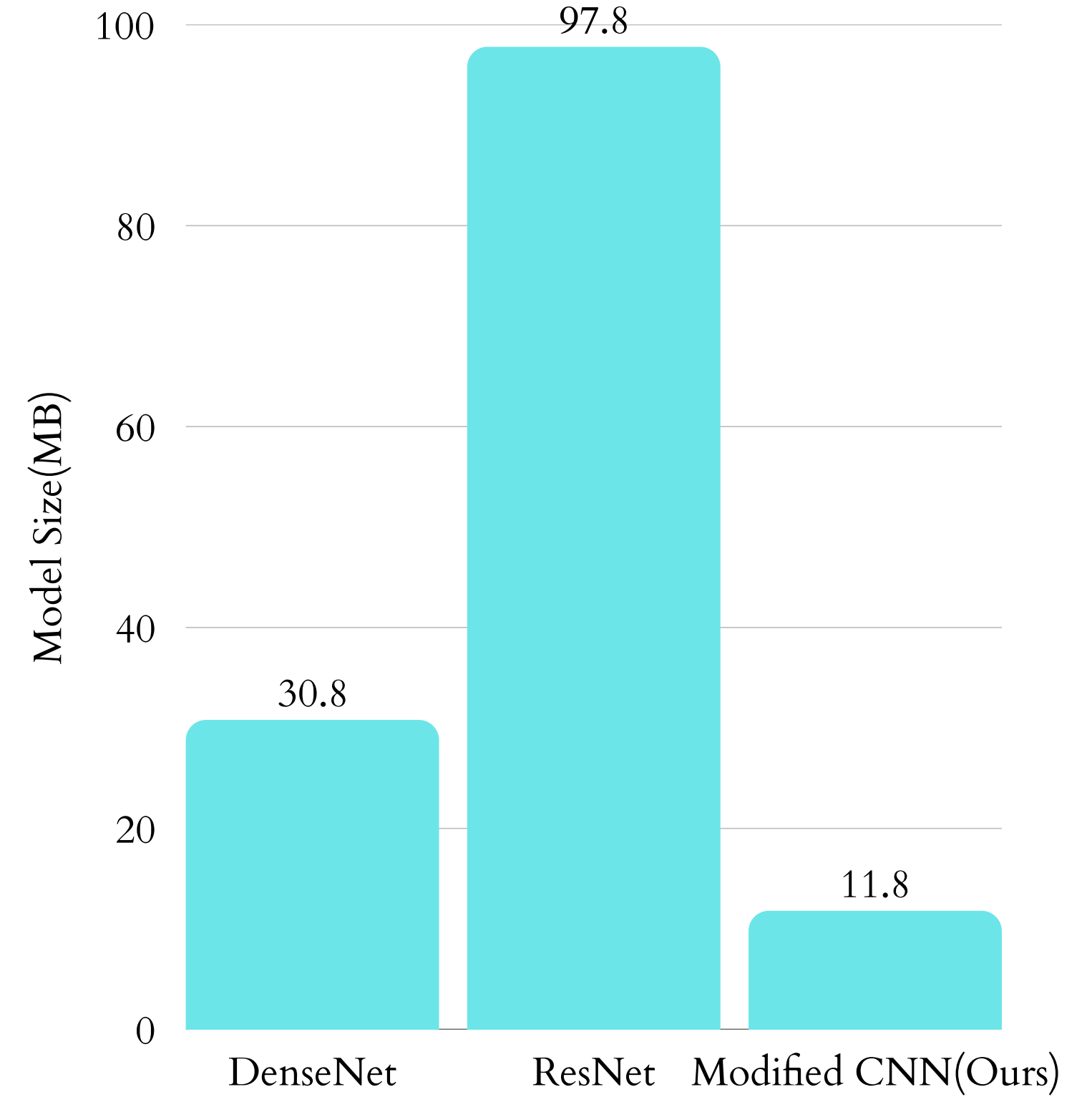
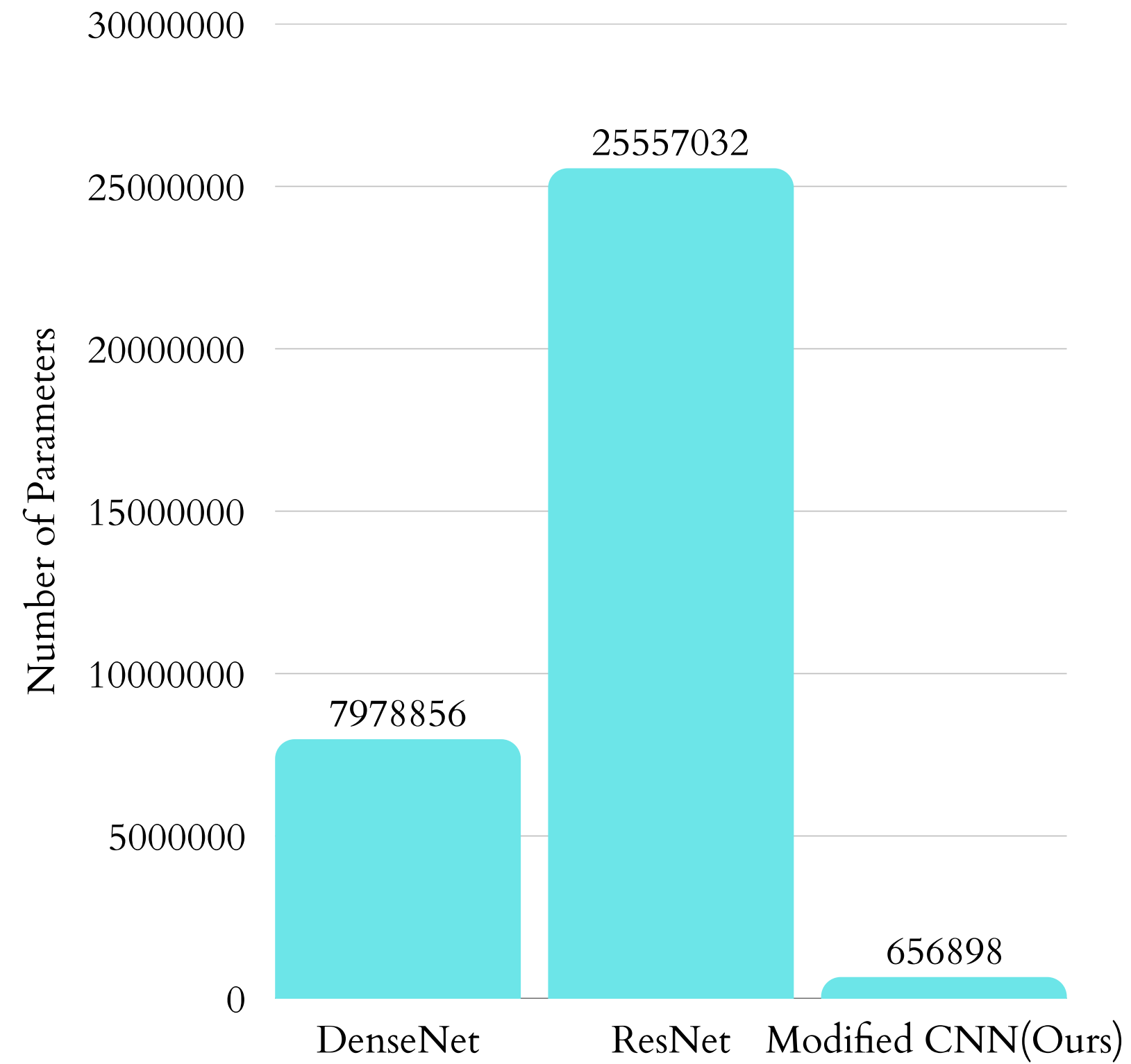
Compare with other models



Generalization Performance (Inference)

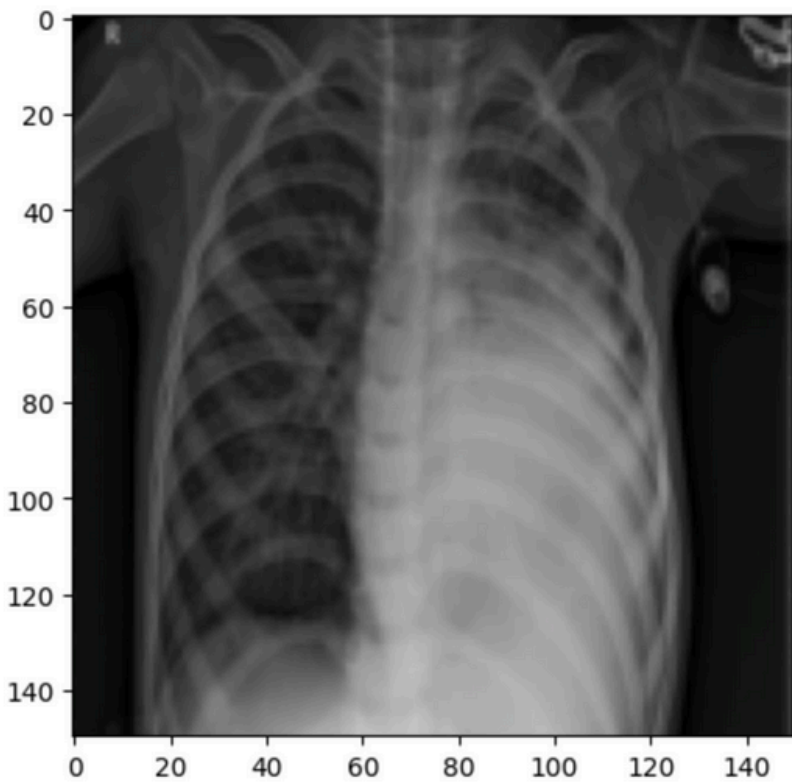


Compare with other models

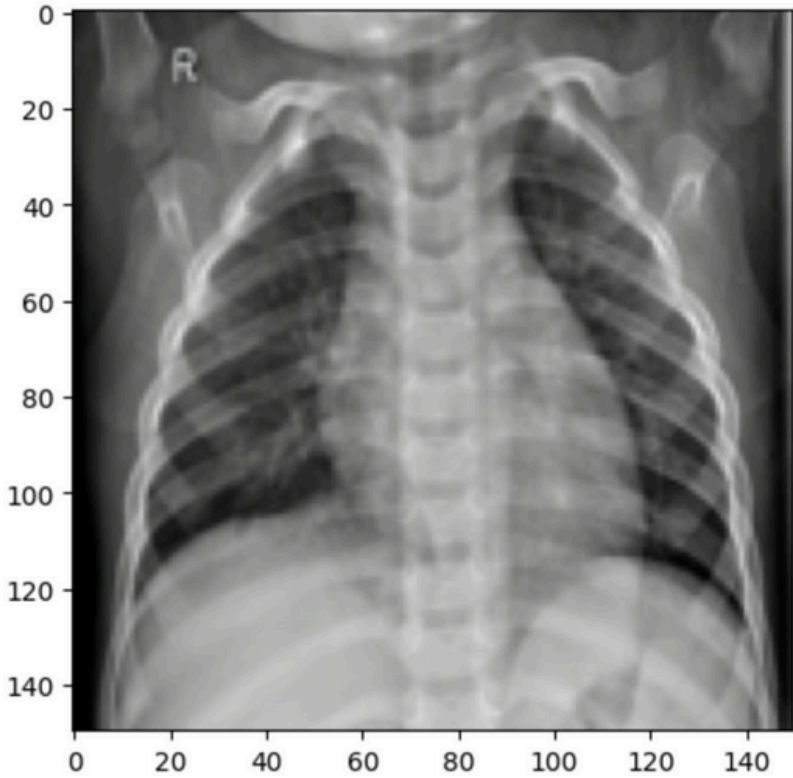


Experiment Result

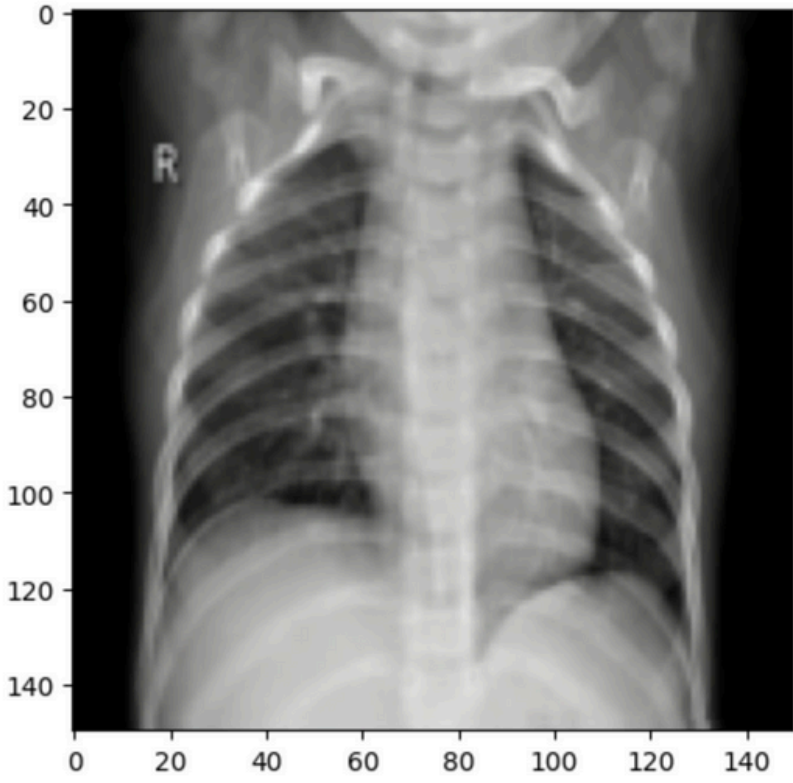
Actual: PNEUMONIA predicted: PNEUMONIA



Actual: NORMAL predicted: NORMAL



Actual: NORMAL predicted: NORMAL



Actual: PNEUMONIA predicted: PNEUMONIA

